

All About Edge Computing Architecture, Open Source Frameworks and IoT Solutions

An IoT solution is an amalgamation of hardware, software and networking capabilities. Edge computing is required for any successful and optimised IoT solution. There are various open source frameworks available for edge computing in the IoT space. This article introduces some of them.



Data is the heart of an IoT system and it needs to be immediately processed. An IoT system can generate limitless data within a second, depending on the various IoT devices assembled under the system, as per business needs. Limitless data generated from IoT devices or sources can easily consume network bandwidth and lead to a need for excess data storage. It is crucial to aggregate and digitise the data at the periphery of the system, which can then be communicated to back-end systems. This responsibility is taken care of by edge computing, and helps to reduce or optimise the IT infrastructure. These edge computing systems reside close to the IoT devices/data sources and also enforce the required security. A major benefit of edge computing is that it improves the time to action, reduces response time and optimises the use of network resources. It also helps to reduce latency and network bottlenecks.

Reference architecture of edge computing for IoT solutions

IoT devices and data sources

An IoT system can have various data sources—IP capable or low-powered devices like sensors, appliances, applications, social media sites, or data from third party systems. These data sources can generate data in various formats, frequencies and volumes. This is the layer where data is getting generated. Data sources may vary as per an enterprise or industry's needs. A solution should support the data source channels required to meet a business' needs. These IoT devices will capture the data and communicate over IoT protocols to a nearby edge gateway system via Wi-Fi, Ethernet, Bluetooth, NFC, Zigbee or any other communication/transport layer protocols.